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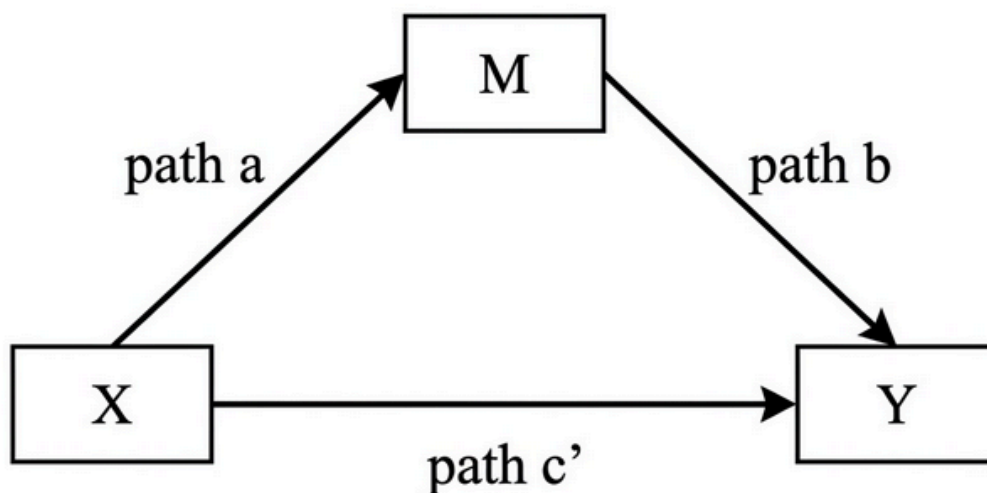
Mediation Analysis: MASTER CONSOLIDATED WORKSHEET

Topic: Understanding Mechanisms, Indirect Effects, and Statistical Interpretation

Instruction: This document combines all essential lecture materials. Follow the slides and complete every section. Do not skip any calculations.

Part 1: Foundations & The Conceptual Core

1. **Generalized Linear Model (GLM):** Mediation is not a separate statistical model; rather, it is an _____ of the GLM.
2. **The Core Question:** While traditional regression focuses on the direct relationship ($X \rightarrow Y$), mediation asks: "What is the _____ that links X to Y ?"
3. **Structural Equation Modeling (SEM):** Mediation is the core of SEM, allowing researchers to build complex _____ of behavior.
4. **Variable Roles:**
 - X is the _____ or Independent variable.
 - M is the _____ or Mechanism.
 - Y is the _____ or Outcome.
5. **Unique Property of M :** In a mediation model, the mediator acts as a _____ variable (predicted by X) and an _____ variable (predicting Y).
6. **Mediation explains the mechanism** of a relationship. It moves from $X \rightarrow Y$ to $X \rightarrow ______ \rightarrow Y$.
7. **The Mediation Triangle:** Study the minimalist diagram below. Label the paths **a**, **b**, and **c'** as defined in the standard model.



8. Thinking about Signs (Example: Food Insecurity & Health)

Consider the relationship between **Food Insecurity** (X), **Health Care Utilization** (M), and **Depressive Symptoms** (Y).

- **Scenario A (Positive Paths):** A researcher finds that higher Food Insecurity (X) leads to more frequent Doctor Visits (M) because of acute health crises (a is positive). These frequent visits lead to more clinical detections of Depressive Symptoms (Y) (b is positive).
- **Scenario B (Negative Paths):** Another researcher finds that higher Food Insecurity (X) makes people *avoid* the doctor to save money (a is negative). The *lack* of health care (M) then causes Depressive Symptoms (Y) to worsen (b is negative).

Questions:

- Calculate the sign of the **Indirect Effect** ($a \times b$) for Scenario A: _____
- Calculate the sign of the **Indirect Effect** ($a \times b$) for Scenario B: _____
- Even though the paths (a and b) have different signs in these scenarios, the resulting indirect effect sign is the same. Explain why this happens mathematically, and what it tells you about the nature of the relationship between Food Insecurity and Depression in both cases:

Part 2: Causal Assumptions & Logic

9. **Causal Sequence:** For mediation to be logically valid, the sequence must be: _____ $\rightarrow$ _____ $\rightarrow$ _____.
10. **Data Limitation:** When using _____ data, establishing causal order can be difficult. You must ensure M logically occurs between X and Y .
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Part 3: The Statistical Model

11. Mediation is estimated using _____ (number) simultaneous regression equations.
12. **Path a :** Represents the effect of _____ on the mediator (_____).
13. **Path b :** Represents the effect of _____ on the outcome (_____), while controlling for X .
14. **Path c' :** Represents the _____ effect of X on Y , while controlling for M .
15. **Indirect Effect:** This is calculated as the product of _____ and _____ ($a \times b$).
16. **Total Effect (c):** The sum of the _____ effect (c') and the _____ effect (ab).
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Part 4: Numeric Problem - The Studying & Stress Study

You are analyzing how **Hours Studied (X)** affects **Test Scores (Y)**, with **Stress Level (M)** as a mediator. You find the following coefficients:

- **Path a** (Effect of X on M): **2.0**
- **Path b** (Effect of M on Y): **-3.0**
- **Path c'** (Direct effect of X on Y): **5.0**

Tasks:

17. Calculate the **Indirect Effect (ab)**: _____ \$
18. Calculate the **Total Effect (c)**: _____ \$
19. **Interpretation:** Does the indirect effect work in the same direction or the opposite direction as the direct effect? _____
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Part 5: Complete vs. Partial Mediation & Confounding

20. **Complete Mediation:** The relationship between X and Y _____ when the mediator is added to the model.

21. **Partial Mediation:** X affects Y both _____ AND through the _____.
22. **Mediator:** Explains a _____ relationship by showing the internal mechanism.
23. **Confounder:** Creates a _____ relationship between X and Y by influencing both.
24. **Suppression Challenge:** Normal mediation makes the $X \rightarrow Y$ effect smaller. When a researcher adds a third variable and the coefficient of original X **increases** and becomes **more significant**, this is called: _____\$

Part 6: Testing & Bootstrapping

25. **Traditional View:** The Baron & Kenny 4-Step method establishes that X and Y must be related for mediation to exist.
26. **Modern View (Hayes):** You **CAN** have mediation without a significant _____ effect. The most important parameter is the _____ effect.
27. **Bootstrapping:** Because indirect effects (ab) are not normally distributed, we use bootstrapping to create a custom **95%** _____ **Interval**.
28. **Significance Check:** If the bootstrapping interval **does not contain** _____, the indirect effect is considered significant.

Part 7: Case Study Exercises (Extraction & Analysis)

A. Example: Education & Income

Model M — Outcome: Job Skills

	coeff	se	t	p
constant	2.0000	.5000	4.0000	.0010
Education (a)	.5000	.1000	5.0000	.0000

29. From the table above, the coefficient for **Path a** is _____.

Model Y — Outcome: Income

	coeff	se	t	p
constant	20000	5000	4.0000	.0010
Education (c')	500	250	2.0000	.0480
Skills (b)	2000	400	5.0000	.0000

30. From the table above, the coefficient for **Path b** is _____.
31. The **Direct Effect** (c') of education on income is _____ ($p =$ _____).
32. **Calculate the Indirect Effect** (ab): _____.

33. **Interpretation:** One additional year of education increases income by _____ dollars specifically through increased job skills.

B. The "Sugar Shortage" Study (PMI)

Below is a partial regression table for **Article Placement** (X), **Presumed Media Influence** (M), and **Intention to Buy** (Y).

Model M — Outcome: PMI

	coeff	se	t	p	LLCI	ULCI
constant	5.3769	.1618	33.2222	.0000	5.0565	5.6973
cond (a)	.4765	.2357	2.0218	.0454	.0099	.9431

34. According to the output above, front-page placement significantly increases PMI by _____ units ($p =$ _____).

Model Y — Outcome: REACTION

	coeff	se	t	p	LLCI	ULCI
constant	.5269	.5497	.9585	.3397	-.5615	1.6152
cond (c')	.2544	.2558	.9943	.3221	-.2522	.7609
pmi (b)	.5064	.0970	5.2185	.0000	.3143	.6986

35. The direct effect (c') of condition on reaction is _____ ($p =$ _____).

36. The effect of PMI (M) on reaction is _____ ($p =$ _____).

37. Is this **Full** or **Partial** mediation based on the c' p -value? _____ \$

Indirect Effect Result

	Effect	BootSE	BootLLCI	BootULCI
PMI	.2413	.1325	.0095	.5304

38. Is the indirect effect significant? _____ (Yes/No). How do you know using the "Zero Rule"? _____.

C. Economic Stress (ESTRESS)

Model M — Outcome: AFFECT

	coeff	se	t	p	LLCI	ULCI
estress (a)	.1729	.0296	5.8308	.0000	.1145	.2313

Model Y — Outcome: WITHDRAW

	coeff	se	t	p	LLCI	ULCI
constant	1.4471	.2520	5.7420	.0000	.9508	1.9433
estress (c')	-.0768	.0524	-1.4667	.1437	-.1800	.0263
affect (b)	.7691	.1031	7.4627	.0000	.5662	.9721

Total Effect (c)

	coeff	se	t	p	LLCI	ULCI
estress (c)	.0561	.0542	1.0353	.3015	-.0506	.1628

39. In the ESTRESS study, path a is _____ and path b is _____.
40. The total effect (c) is _____ ($p =$ _____).
41. In this study, the **Total Effect was NOT significant**, but the **Indirect Effect WAS significant**.
According to modern theory, does mediation still exist? _____ \$
42. Why did the total effect "disappear" despite a significant indirect effect?
_____.

D. Case Study - The Legal Protest Study

Scenario Details:

- **Protest (X):** A lawyer protests against gender discrimination (0 = no, 1 = yes)
- **Respappr (M):** Perceived adequacy of response (scale 1-7)
- **Like (Y):** Assessment of the lawyer (scale 1-7)

1. **State the Hypothesis:** State the formal hypothesis (the predicted relationship between X , M , and Y):

2. **The Regression Equations:** Using the results from the table below, write the full regression equations:

- $M =$ _____ $+$ _____ $X + e_M$
- $Y =$ _____ $+$ _____ $X +$ _____ $M + e_Y$

3. Summary of Model Coefficients:

Antecedent		Consequent: respappr(Mi)					Consequent: liking(Y)			
		Coef	SE	t	p		Coef	SE	t	p
protest(X)	a	0.798	0.128	6.248	<.001	c'	-0.118	0.113	-1.044	.298
respappr(Mi)						b	0.420	0.069	6.114	<.001
Constant	i	4.043	0.168	24.077	<.001	i	3.716	0.306	12.125	<.001

4. Testing & Verification:

- Calculate the **Indirect Effect (ab):** _____

- Verify the additive property (Total = Direct + Indirect): _____ + _____ = _____

5. Interpretation & Analysis:

- Interpret the R^2 for **liking** (Y) (Assume $R^2 = 0.251$):
_____ \$
- Does the data support the hypothesis? Explain why:
_____ \$

Part 8: Practical Software, Conventions, & Drawing

43. In **SPSS** or **R**, we primarily use the _____ Macro developed by Andrew Hayes.
44. To test simple mediation (one X , one M , one Y), we use **Model** _____ in the macro.
45. Most research literature uses _____ (Standardized / Unstandardized) coefficients for mediation results.

Drawing Tasks: 46. Draw the **Conceptual Diagram** for the Legal Protest study (labeled with variable names). 47. Draw the **Statistical Diagram** and fill in the paths with coefficients (a , b , and c') found in the tables above.